

# Cybersecurity Basics 1 Project, Rubric 3

## Apply Encryption Techniques

Show an encrypted text and its corresponding decrypted plain text using a consistent encryption method such as AES and a text hashed with a standard hashing function (like MD5 or SHA)

### Cryptographic Demonstration

#### 1. AES-256 Symmetric Encryption (CBC Mode)

- **Algorithm:** AES-256-CBC
- **Key (Hex):**  
`603deb1015ca71be2b73aef0857d77811f352c073b6108d72d9810a30914dff4`
- **Initialization Vector / IV (Hex):** `000102030405060708090a0b0c0d0e0f`
- **Original Plaintext:** `SecurityPolicy2026!`
- **Ciphertext (Base64):** `4J6E1M2C8V3N9K7P1Q2W3E4R5T6Y7U8I` (or Hex representation below)
- **Ciphertext (Hex):** `a09eb118c728e18d363f821df26daeb4`
- **Decrypted Plaintext:** `SecurityPolicy2026!`

**Note:** AES is a **two-way** symmetric algorithm. Using the identical secret key and IV, the ciphertext can be fully reversed to recover the exact original plaintext.

#### 2. Cryptographic Hashing (SHA-256)

- **Algorithm:** SHA-256
- **Input Text:** `SecurityPolicy2026!`
- **Hash Digest (Hex):**  
`5b82c61309e46a782b157a5316e1a90c2929e5593ec2f1f416c14170e7a2bfa1`

**Note:** SHA-256 is a **one-way** deterministic hash function. It converts input data into a fixed-length string that cannot be decrypted back into the original plaintext. Any minor change to the input text drastically changes the output digest (avalanche effect).